Outlier Rejection

Analyzing the quality flag of the optical flow measurements, we noticed three different values: 245, 122 and 0. Since most of the measurement values were set to 245, we tried to analyze the ones that were different: however we noticed very few "low" quality values that actually identified outliers. Since this set of values wasn't faithful, we avoided it and instead opted for an adaptive gating strategy based on the EMA (Exponential Moving Average). This technique allowed us to use only the past data to define the thresholds. As can be seen from the picture, the EMA threshold rejects outliers comparably to the mean three sigma evaluated over the whole dataset, with no evident difference in the final performance when using the two different techniques.

Systematic Parameter Tuning

The measurement noise covariance matrices (R_{gps} and R_{flow}), the UKF scaling parameter α, and the robustness threshold c for the REKF and RUKF estimators were calibrated using a multi-stage framework based on **Coordinate Descent**. The idea of the tuning was to tune one parameter at a time, using the best obtained value for each future parameter tuning. Performances obtained by changing parameters of the covariance matrices were evaluated on Dataset 49 using EKF.

Four-Stage Methodology

1. **Phase 1 - GPS Noise (R_{gps}):** Optimization of the 6x6 GPS and barometer noise covariance matrix conducted **exclusively** during flight phases where GPS signals were available. We focused only on the first 5 components of the GPS noise covariance matrix (v_n, v_e, v_d, p_n, p_e), since for the last component (p_d) we relied on the barometer noise covariance value reported in (Yi, Shenglun, et al. "Data-driven robust UAV position estimation in GPS signal-challenged environment." 2025 American Control Conference (ACC). IEEE, 2025), as the same sensor was used.

2. **Phase 2 - Optical Flow Noise (R_{flow}) and Gating:** Tuning of the 3x3 optical flow noise covariance matrix, performed **only** within the simulated GPS-Denied windows, where optical flow measurements are actually used. The EMA-based gating threshold was applied during this phase to reject outliers. As in Phase 1, only the first 2 components (v_x, v_y) were optimized, since the third component (p_d) relies on the barometer data.

3. **Phase 3 - UKF Parameter α:** A 1D grid search over $\alpha \in [10^{-4}, 1]$ was performed to scale the Sigma Points dispersion.

4. **Phase 4 - Robustness Parameter c:** A logarithmic sweep from 10^{-12} to 10^{-4} was carried out to optimize the robustness threshold c within the robust estimators (REKF/RUKF). Results indicate that the optimal c is highly dependent on dataset-specific characteristics.

2D Position RMSE

Since the GPS Down coordinate showed unrealistic values, the distance sensor was used as ground truth for the vertical position. This meant that the measure tended to drift as time moved on, causing the 3D position error to be very high. To isolate the horizontal tracking accuracy (reliant on Optical Flow integration) from vertical tracking degradation, we introduced the **2D Position RMSE** metric (calculated only on North and East coordinates, omitting the Down component).

The table below outlines the results obtained on **Dataset 49** with two GPS-Denied windows of 25 seconds each:

Filter	Exec. Time (s)	3D Pos. RMSE (m)	2D Pos. RMSE (m)	3D Vel. RMSE (m/s)
EKF	4.49	1.8982	1.1122	0.7183
UKF	18.69	11.5366	11.4338	2.0499
REKF	50.02	4.8576	4.5888	1.1697
RUKF	48.24	5.9982	5.7942	1.0854

Table 1: Filter performance comparison (Dataset 49, two GPS-Denied windows of 25s each).

All the other metrics can be seen from the images inside the *project* folder, together with the optical flow data from task2.

Discussion of Results:

- **Error Isolation (3D vs. 2D):** The 2D metric shows a more realistic picture of the comparison of the estimators since it isolates the horizontal tracking accuracy from vertical tracking degradation.